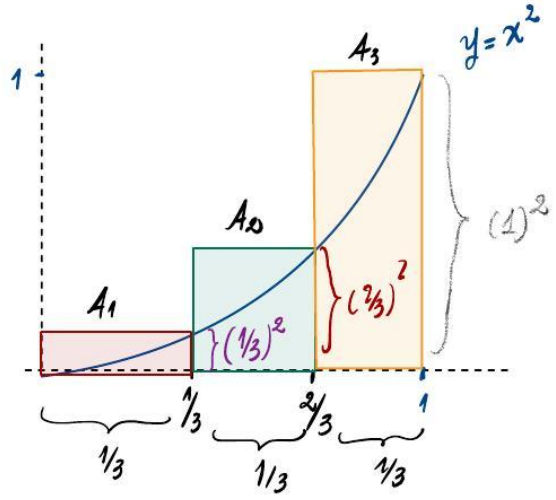


$$A \approx A_1 + A_2 + A_3$$

$$A_1 = \left(\tfrac{1}{3}\right)\left(\tfrac{1}{3}\right)^2$$

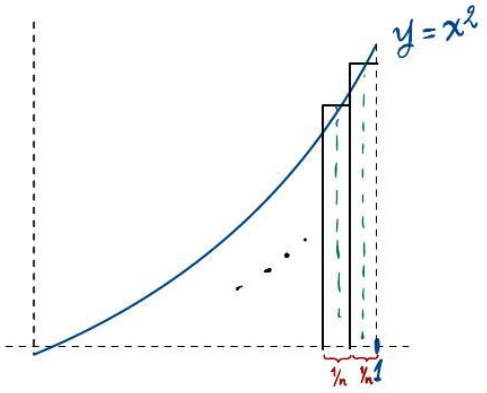
$$A_2 = \left(\tfrac{1}{3}\right)\left(\tfrac{2}{3}\right)^2$$

$$A_3 = \left(\tfrac{1}{3}\right)(1)^2$$

$$A \approx \frac{1}{3^3} + \frac{4}{3^3} + \frac{1}{3}$$

Using geogebra we can check that $A = \frac{1}{3}$.

Example. Show that the area in the previous example is $1/3$.



$$A_n = \frac{1}{n}\left(\frac{1}{n}\right)^2 + \frac{1}{n}\left(\frac{2}{n}\right)^2 + \cdots + \frac{1}{n}\left(\frac{n}{n}\right)^2$$

$$= \frac{1}{n}\left\{\frac{1}{n^2} + \frac{2^2}{n^2} + \cdots + \frac{n^2}{n^2}\right\}$$

$$= \frac{1}{n^3}\left\{1 + 2^2 + 3^2 + \cdots + n^2\right\}$$

$$= \frac{1}{n^3} \cdot \sum_{i=1}^{n} i^2$$

Since $\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$ So,

$$A_n = \frac{1}{n^3} \cdot \frac{n(n+1)(2n+1)}{6}$$

this,

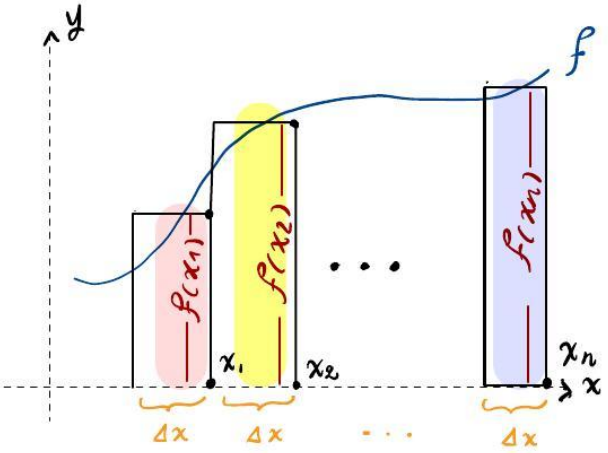
$$A = \lim_{n\to\infty} A_n = \lim_{n\to\infty} \frac{1}{n^3} \frac{n(n+1)(2n+1)}{6}$$

$$= \lim_{n\to\infty} \frac{1}{n^3} \cdot n \cdot n(1+1/n)\,n(2+1/n) \Big/ 6$$

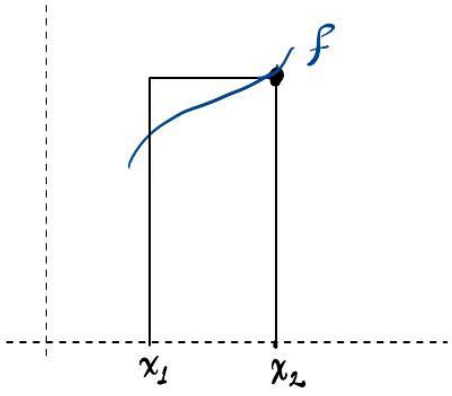
$$= \lim_{n\to\infty} \frac{(1+1/n)(2+1/n)}{6} = \frac{2}{6} = \frac{1}{3}.$$

Def. The area A of the region that lies under the curve of a continuous function f is the limit of the areas of approximating rectangles.

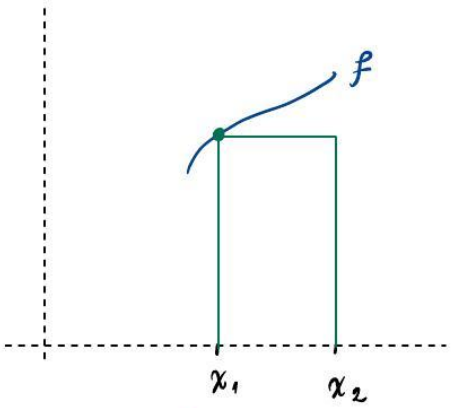
$$A = \lim_{n\to\infty} A_n = \lim_{n\to\infty} \{ f(x_1)\Delta x + f(x_2)\Delta x + \cdots + f(x_n)\Delta x \}$$



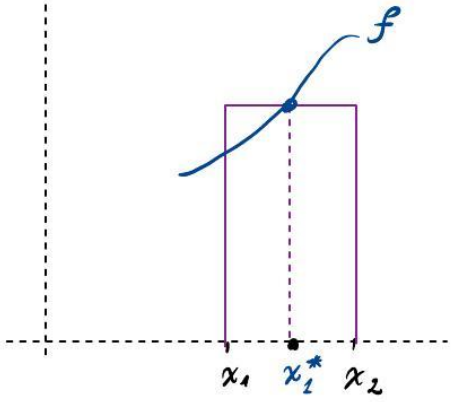
Upper-Sum



Upper-Sum



Low-Sum



Riemann-Sum

In the Riemann sence we have:

$$A = \lim_{n\to\infty} \{ f(x_1^*)\Delta x + f(x_2^*) + \cdots + f(x_n^*)\Delta x \}$$

Example. Let A be the area of the region that lies under the graph of $f(x) = \cos x$ between $x=0$ and $x=b$ where $0 \le b \le \frac{\pi}{2}$.

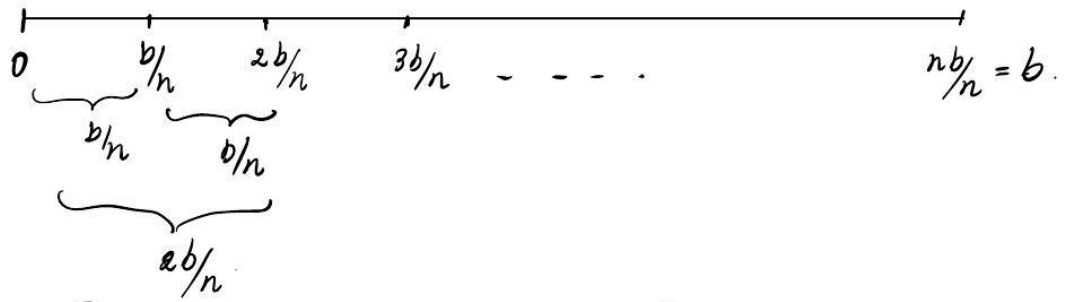
a) Using the right end points find the expression for A as a limit. Do not evaluate that limit.

b) Estimate the area for $b = \frac{\pi}{2}$ by taking the sample points to be midpoints.

Sln: a). Since $a=0$ the width of the rectangles are:

$$\Delta x = \frac{b-a}{n} = \frac{b}{n}.$$

So, $x_1 = \frac{b}{n}$, $x_2 = \frac{2b}{n}$, $x_3 = \frac{3b}{n}$ - - - - $x_n = \frac{nb}{n}$.



$$A_n = f(x_1)\Delta x + f(x_2)\Delta x + \cdots + f(x_n)\Delta x$$

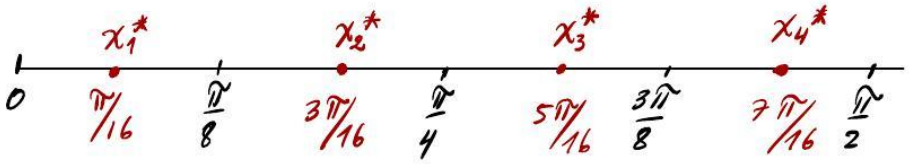
$$= \cos x_1 \cdot \Delta x + \cos x_2 \Delta x + \cdots + \cos x_n \Delta x$$

$$= \frac{b}{n}\left\{\cos\left(\frac{b}{n}\right) + \cos\left(\frac{2b}{n}\right) + \cos\left(\frac{3b}{n}\right) + \cdots + \cos(b)\right\}$$

$$= \frac{b}{n}\sum_{i=1}^{n}\cos\left(\frac{bi}{n}\right)$$

$$A = \lim_{n\to\infty}\frac{b}{n}\sum_{i=1}^{n}\cos\left(\frac{bi}{n}\right).$$

b). with $n=4$ and $b=\frac{\pi}{2}$. $\Delta x = \frac{\frac{\pi}{2}-0}{4} = \frac{\pi}{8}$ then.



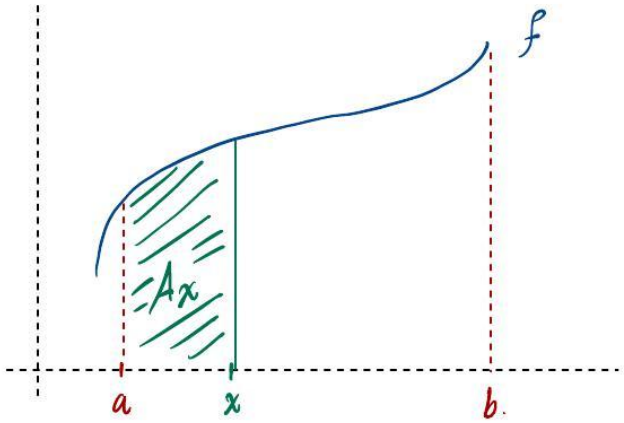
$$R_4 = \sum_{i=1}^{4} f(x_i^*)\Delta x = \Delta x \cdot \sum_{i=1}^{4} f(x_i^*)$$

$$= \frac{\pi}{8}\left(\cos\left(\frac{\pi}{16}\right) + \cos\left(\frac{3\pi}{16}\right) + \cos\left(\frac{5\pi}{16}\right) + \cos\left(\frac{7\pi}{16}\right)\right) \approx 1.006.$$

The fundamental Theorem of Calculus.

* The fundamental theorem of Calculus part 1.

The first part of the FT deals with the function defined by an equation.



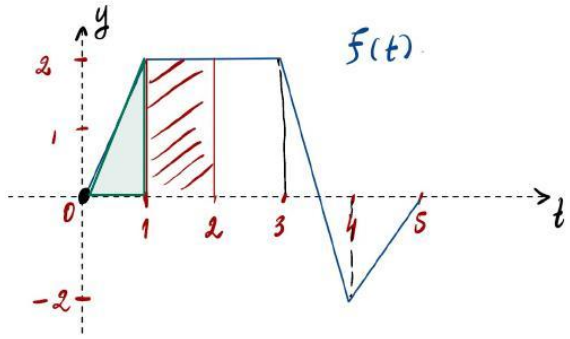
Then, notice that this area depends on x.

$$g(x) = \int_a^x f(t)\,dt$$ ⟶ The integral of $f(t)$ between a and x.

area under the curve $f(t)$ between a and x.

Where f is a continuous function on $[a,b]$ and $a \le x \le b$. Observe that g depends only on x. Which appears as the variable upper limit in the integral.

Example.



If $g(x) = \int_0^x f(t)\,dt$.

find the values $g(0)$, $g(1)$, $g(2)$, $g(3)$, $g(4)$, $g(5)$.

Sln.

$$g(0) = \int_0^0 f(t)\,dt = 0$$

$$g(1) = \int_0^1 f(t)\,dt = \frac{1}{2}(1)(2) = 1.$$

$$g(2) = \int_0^2 f(t)\,dt = \int_0^1 f(t)\,dt + \int_1^2 f(t)\,dt = 1 + (1)(2) = 1+2 = 3.$$

$g(3) = 3+2 = 5$ $\quad$ $g(4) = ?$ $\quad$ $g(5) = 5.$

The Fundamental Theorem of Calculus Part 1

If f is a continuous function on $[a,b]$ then the function g defined by

$$g(x)=\int_a^x f(t)\,dt \quad ; \quad a\le x\le b$$

is continuous on $[a,b]$ and differentiable on (a,b) and

$$g'(x)=f(x).$$

Example. find the derivative of $g(x)=\int_0^x \sqrt{1+t^2}\,dt$.

Sln. Since $f(t)=\sqrt{1+t^2}$ so

$$g(x)=\int_0^x f(t)\,dt.$$

So by FTCP1 $\quad g'(x)=f(x)=\sqrt{1+x^2}$.

Example. Find $\dfrac{d}{dx}\int_1^{x^4} \sec t\,dt$.

Sln: let $f(t)=\sec t$ but

$$g(x)=\int_1^{x^4} \sec t\,dt = h(s(x))$$

where $h(x)=\int_1^x f(t)\,dt$ and $s(x)=x^4$

$$h(s(x))=h(x^4)=\int_1^{x^4} f(t)\,dt = g(x) \quad \checkmark$$

$$g'(x)=\underset{c'R}{h'(s(x))\cdot s'(x)} = f(s(x))\cdot s'(x) = 4x^3\sec(x^4)$$

$$g(x)=\int_1^{x^4}\sec t\,dt \Rightarrow \quad g'(x)=\sec(x^4)\cdot\frac{d}{dx}(x^4)=4x^3\sec(x^4).$$

Problem for you! Compute $\dfrac{d}{dx}\int_{\sqrt{x}-x}^{x^2+4x+1}\cos(t)\,dt$.

Properties: i) $\int_a^a f(x)\,dx=0$

ii) $\int_a^b f(x)\,dx=-\int_b^a f(x)\,dx$.

The Fundamental Theorem of Calculus Part 2.

If f is a continuous function on $[a,b]$ then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

where F is any antiderivative of f, i.e. F is s.t. $F' = f$.

Example. Evaluate the integral $\int_{-2}^{1} x^3\,dx$.

Sln. We need an antiderivative for $f(x) = x^3$; $F(x) = \frac{x^4}{4}$.

$$\int_{-2}^{1} x^3\,dx = F(1) - F(-2) = \frac{1^4}{4} - \frac{(-2)^4}{4} = \frac{1}{4} - 4 = \frac{-15}{4}.$$

$F_2 = \frac{x^4}{4} + \pi$

$$\int_{-2}^{1} x^3\,dx = F_2(1) - F_2(-2) = \left(\frac{1^4}{4} + \pi\right) - \left(\frac{(-2)^4}{4} + \pi\right)$$

$$= \frac{1^4}{4} + \cancel{\pi} - \frac{(-2)^4}{4} - \cancel{\pi}$$

$$= -\frac{15}{4}.$$

Example. Find the area under the parabola $y = x^2$ for $0 \le x \le 1$.

Sln. $A = \int_0^1 x^2\,dx$ So, $f(x) = x^2$ $F(x) = \frac{x^3}{3}$. by FTCP2.

$$A = F(1) - F(0) = \frac{1}{3} - 0 = \frac{1}{3}.$$

Example. Find the area under the curve of cosine curve from 0 to b where $0 \le b \le \frac{\pi}{2}$.

Sln.

$$A = \int_0^b \cos x\,dx = \sin x \Big|_0^b = \sin b - \sin 0 = \underline{\sin b}$$

In particular if $b = \frac{\pi}{2}$ then

$$A = \int_0^{\pi/2} \cos x\,dx = \sin\frac{\pi}{2} = 1.$$

Problem: What is wrong with the following calculation?

$$\int_{-1}^{3} \frac{1}{x^2}\,dx = \frac{x^{-1}}{-1}\Bigg|_{-1}^{3} = -\frac{1}{3} - 1 = -\frac{4}{3}.$$